

I seek admission to the Finance PhD program at Olin Business School. My research centers on how incentives, information, and contractual design shape decision-making in corporate, credit, labor, and innovation contexts. I am drawn to corporate finance because firms—core units of modern economies—sit at the intersection of innovation, social welfare, and the complex dynamics among agents. I use causal inference, structured modeling, and data design to connect theory with evidence and expand what can be measured in finance. Olin’s data-driven culture, where I have already worked closely with faculty and fellow students, makes it the perfect place for my research training.

My professional and academic experiences laid the groundwork for my motivation to study finance through research. During my internship experiences, I observed how firms design financing strategies, allocate risk, and navigate contracting frictions. These experiences intrigued me to understand the mechanisms behind those decisions rather than execute them. My senior thesis and early projects in green finance introduced me to empirical work but lacked the methodological depth needed for credible inference, motivating me to study identification and causal design more seriously. Later, a project on detecting LLM text authorship exposed me to language models as powerful tools for analyzing unstructured data and broadened my view of what can be measured in finance. Together these experiences shaped my transition from practice to inquiry and motivated me to pursue formal research training at Olin.

My work with Professor Margarita Tsoutsoura centered on empirical corporate finance studies that drew on firm, worker, and policy data from diverse and often unconventional sources. I developed new ways to collect and link information—building robust scrapers, designing matching mechanisms, and creating validation routines to ensure coverage and accuracy. When key measures did not exist, I built them from scratch—a few examples include measures of industry skilled-worker intensity, family-office career paths, and trade-basket unit values. Working across these projects made me comfortable handling large-scale structured and unstructured data and taught me to view data construction as part of research design itself. I came to see that credible empirical work depends on careful construction, clear documentation, and sound measurement, which motivates me to strengthen my econometric and identification skills in doctoral study.

I worked with Professor Brett Green on research examining how deal contingencies (e.g., due diligence, exclusivity, and financing clauses) shape offer valuation and bidding outcomes in takeovers. I designed a local small-LLM and LoRA system to improve and ultimately replace the manual data-collection process, automating extraction and organizing information into bidder-level datasets. The workflow standardizes the “Background of the Merger” sections and can handle both existing and new deals efficiently, with most additional work running in the background when new contingencies or contractual features are added. Building it meant addressing language-model issues such as hallucination, context limits, and document consistency while keeping paragraph-level traceability for validation. It reshaped how I approach research design and opened new questions that become testable through this unique data framework.

Under the guidance of Professor Armando Gomes, I worked on dynamic models of a unique credit product and durable-goods choice to study how incentive and participation constraints shape household decisions. I began by rebuilding an existing two-period framework, then extended it into a full infinite-horizon dynamic-programming model with mixed discrete and continuous choices. The model was

implemented in Python with value- and policy-iteration matrix solvers, using just-in-time compilation to achieve stable convergence and efficient global search. I refined the algorithms through months of iterations, improving robustness and scalability across parameters. Throughout this experience I built a solid foundation in numerical optimization and dynamic programming and motivated me to explore these methods in greater depth during PhD training. It also fueled my interest in combining structured modeling with empirical testing, linking theoretical design to observable financial behavior.

Beyond these research experience, I completed three PhD-level courses at Olin to strengthen the theoretical base of my research training. In *Introduction to Corporate Finance* instructed by Professor Armando Gomes, I ranked first in a cohort also containing then-first-year PhD students. The course gave me a clearer view of how incentive and market frictions drive firm behavior - the mechanisms central to my research interests. *Microeconomic Theory* and *Asset Pricing* further enhanced my analytical foundation and quantitative reasoning. Together, these courses reinforced my motivation to build a rigorous foundation in finance and affirmed my readiness for doctoral-level research training.

I have also served as a teaching assistant for both graduate and undergraduate courses at Olin. Working with students from different backgrounds taught me to explain complex ideas with clarity and adapt them to different ways of thinking. I found teaching deeply rewarding—it helped me think more clearly and connect abstract concepts to intuition. It also strengthened my ability to communicate precisely, both in writing and in discussion. I see research and pedagogy as complementary parts of academic life and view teaching not as a secondary duty but as an integral part of my future career in academia.

My path so far has shaped how I work as a researcher. I value independence, creativity, and persistence—whether designing new ways to measure data, building systems from scratch, or refining models until they are stable. I am comfortable operating at a doctoral level and enjoy collaboration as much as individual problem-solving. Teaching has also demonstrated my communication skills and my ability to translate complex ideas into clear reasoning. One area I aim to improve is overcoming perfectionism. I tend to over-engineer details, but I am learning to view research as an iterative process that rewards steady progress and reflection. I also look forward to rigorous training in empirical identification and causal methods during the PhD to strengthen the analytical side of my work.

Olin is where I want to continue this path. Faculty members here combine data-driven principles with a rigorous approach, which aligns with how I think and work. I look forward to continuing my learning and collaboration with the faculty and exceptional PhD students who have already shaped how I understand good research. The PhD program will allow me to extend current projects into new questions while developing the tools to turn ideas into rigorous, publishable work. In the long run, I hope to build a career that joins research and teaching to advance credible, data-grounded understanding of financial behavior.